Product Specification

DLC Display Co., Limited

DLC-T1.3CDS#1

Features

240(RGB) X 240

IPS/SPI Display

Without Touch

Ph: 86-755-86029824
Fx: 86-755-86029827
www.dlcdisplay.com
sales@dlcdisplay.com

Table of contents

Document Revision History 2
General Description 3
Mechanical Drawing 4
PIN assignment 5
Absolute maximum Ratings 6
Electrical Specifications 6
Command/AC Timing 7
Optical Specification 9
Environmental / Reliability Tests 10

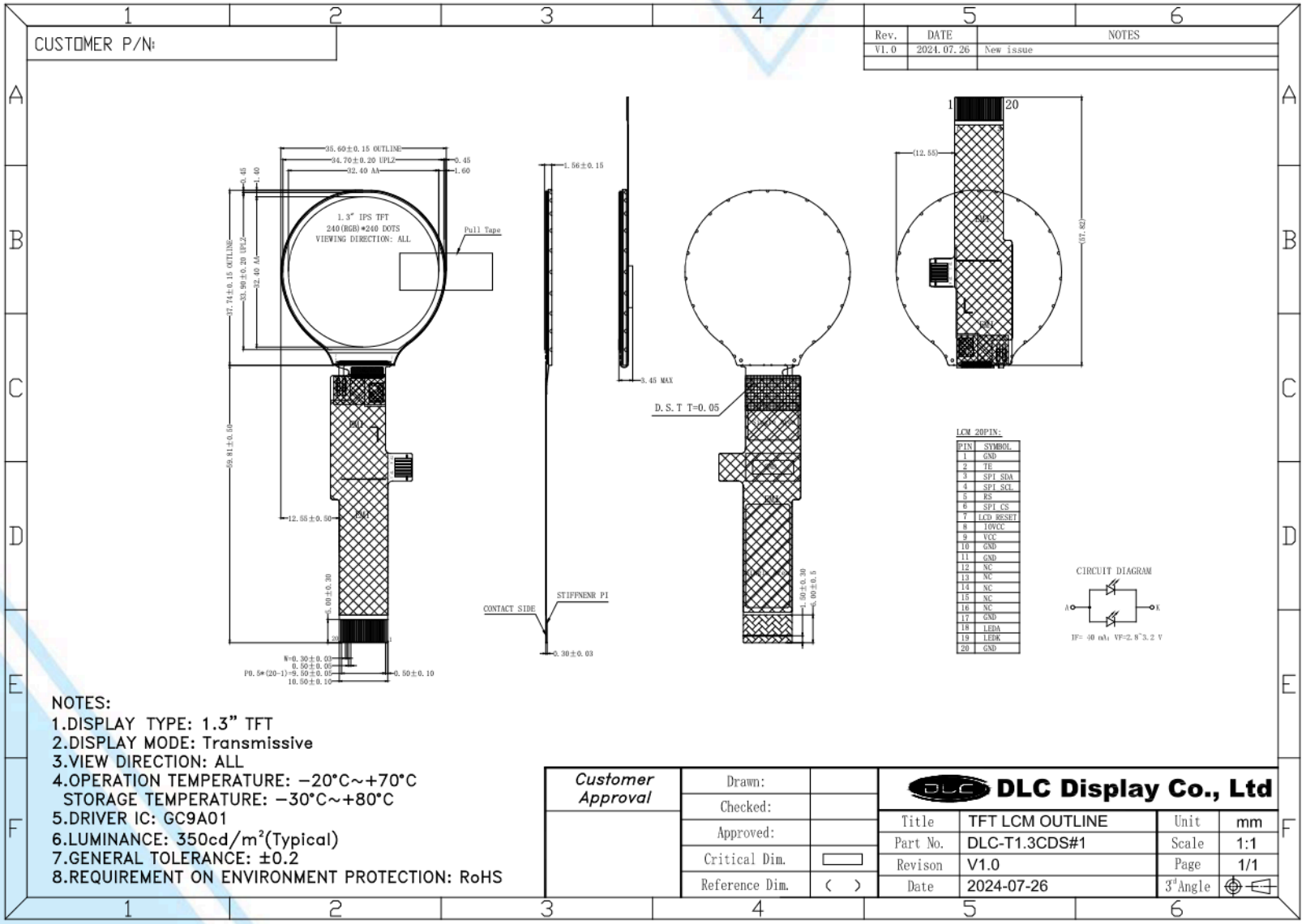
Document Revision History

Date	Revision#	Description
2024-07-26	1.0	Rev 1.0 was issued

General Description

Item	Contents	Unit
Size	1.3	inch
Resolution	240(RGB) x 240	/
Interface	SPI	/
Technology type	IPS	/
Pixel pitch	0.135 x 0.135	mm
Pixel Configuration	R.G.B. Vertical Stripe	
Outline Dimension (W x H x D)	35.60 x 37.74 x 1.56	mm
Active Area	32.40 x 32.40	mm
Display Mode	Transmissive, Normally Black	/
Display Driver IC	GC9A01	/
Viewing Direction	ALL	/
Backlight Type	LED	/

Mechanical Drawing



PIN assignment

No	Symbol	Description	Remarks
1	GND	Ground	
2	TE	Tearing effect signal is used to synchronize MCU to frame memory	
3	SPI_SDA	Serial data input	
4	SPI_SCL	Serial clock input	
5	RS	Serves as command or parameter select	
6	SPI_CS	Chip selection pin	
7	LCD_RESET	This signal will reset the device and must be applied to properly initialize the chip.	
8	IOVCC	Power supply voltage for logic circuit	
9	VCC	Power supply voltage for analog circuit	
10~11	GND	Ground	
12~16	NC	No connection	
17	GND	Ground	
18	LEDA	LED backlight Anode	
19	LEDK	LED backlight Cathode	
20	GND	Ground	

Absolute maximum Ratings

- Electrical Absolute max. ratings

Parameter	Symbol	MIN	MAX	Unit	Remark
Power Supply Voltage	VCC	-0.3	4.6	V	Ta=25° C
	IOVCC	-0.3	4.6	V	

- Environment Conditions

Item	Symbol	MIN	MAX	Unit	Remark
Operating Temperature	TOPR	-20	70	℃	
Storage Temperature	TSTG	-30	80	℃	

Electrical Specifications

- Electrical characteristics

Item		Symbol	MIN	TYP	MAX	Unit	Remark
Power Supply Voltage		VCC	2.5	2.8	3.3	V	
		IOVCC	1.65	2.8	3.3	V	
Power Supply Current		ILCD	-	8	12	mA	
Input voltage	"H"	VIH	0.7*IOVCC	-	IOVCC	V	
	"L"	VIL	0	-	0.3*IOVCC	V	

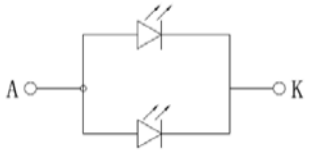
- LED Backlight

Item	Symbol	MIN	TYP	MAX	Unit	Remark
Forward Current	IF	-	40	-	mA	Note1
Forward Voltage	VF	2.8	3.0	3.2	V	
LED Life time	-	30,000	-	-	Hr	Note2

Notes:

1. The LED Supply Voltage is defined by the number of LED at Ta=25℃ and IL =40mA.
2. The "LED life time" is defined as the module brightness decrease to 50% original brightness at Ta=25℃ and IL =40mA. The LED lifetime could be decreased if operating IL is larger than 40mA.

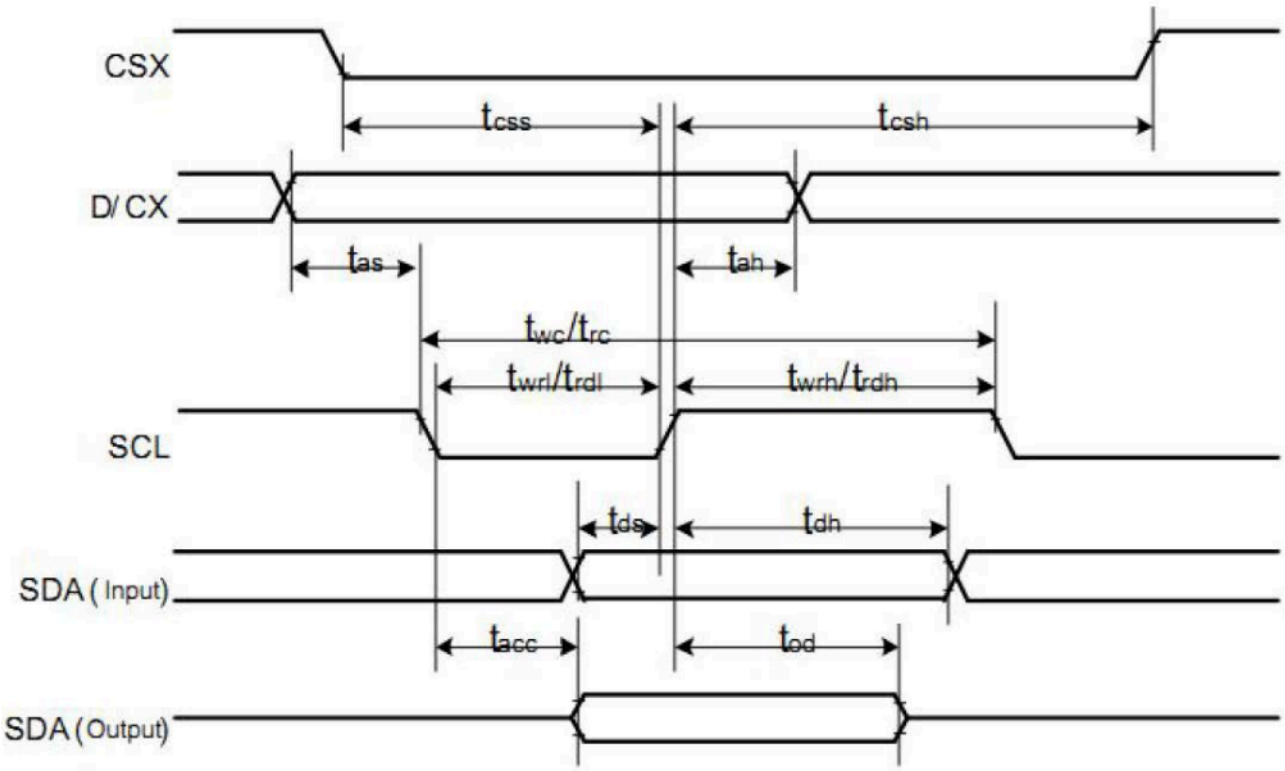
CIRCUIT DIAGRAM



IF= 40 mA; VF=2.8~3.2 V

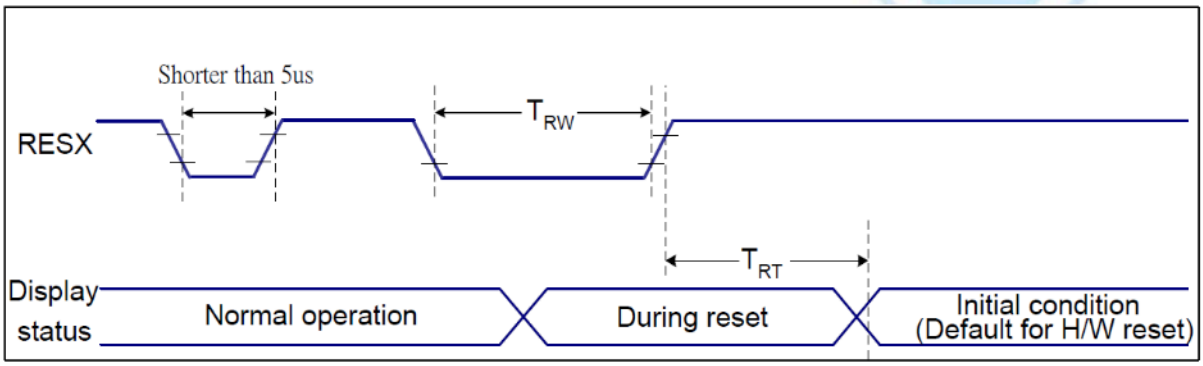
Command/AC Timing

- Serial Interface Characteristics (4-line serial)



Signal	Symbol	Parameter	Min.	Max.	Unit	Description
CSX	tcss	Chip select time (Write)	20	-	ns	
	tcsh	Chip select hold time (Read)	40	-	ns	
SCL	twc	Serial Clock Cycle (Write)	10	-	ns	
	twrh	SCL "H" Pulse Width (Write)	5	-	ns	
	twrl	SCL "L" Pulse Width (Write)	5	-	ns	
	trc	Serial Clock Cycle (Read)	150	-	ns	
	trdh	SCL "H" Pulse Width (Read)	60	-	ns	
	trdl	SCL "L" Pulse Width (Read)	60	-	ns	
D/CX	tas	D/CX setup time	10	-	ns	
	tah	D/CX hold time (Write/Read)	10	-	ns	
SDA/SDI (Input)	tds	Data setup time (Write)	5	-	ns	
	tdh	Data hold time (Write)	5	-	ns	
SDA/SDO (Output)	tacc	Access time (Read)	10	-	ns	

● Reset Timing



VDDI=1.65 to 3.3V, VDD=2.4 to 3.3V, AGND=DGND=0V, Ta=25℃

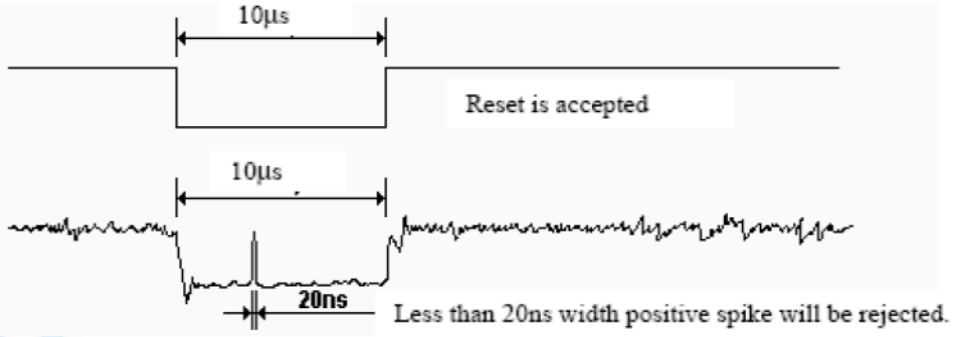
Related Pins	Symbol	Parameter	MIN	MAX	Unit
RESX	TRW	Reset pulse duration	10	-	us
	TRT	Reset cancel	-	5 (Note 1, 5)	ms
				120 (Note 1, 6, 7)	ms

Notes:

1. The reset cancel includes also required time for loading ID bytes, VCOM setting and other settings from NVM (or similar device) to registers. This loading is done every time when there is HW reset cancel time (tRT) within 5 ms after a rising edge of RESX.
2. Spike due to an electrostatic discharge on RESX line does not cause irregular system reset according to the table below:

RESX Pulse	Action
Shorter than 5us	Reset Rejected
Longer than 9us	Reset
Between 5us and 9us	Reset starts

3. During the Resetting period, the display will be blanked (The display is entering blanking sequence, which maximum time is 120 ms, when Reset Starts in Sleep Out –mode. The display remains the blank state in Sleep In –mode.) and then return to Default condition for Hardware Reset.
4. Spike Rejection also applies during a valid reset pulse as shown below:



5. When Reset applied during Sleep In Mode.
6. When Reset applied during Sleep Out Mode.
7. It is necessary to wait 5msec after releasing RESX before sending commands. Also Sleep Out command cannot be sent for 120msec.

For more details, please refer to IC datasheet: IC Datasheet

Optical Specification

Item		Symbol	Condition	Min	Typ.	Max.	Unit	Remark
Contrast Ratio		CR	θ=0°	900	1100	-		Note1 Note2
Response Time		TR+TF	25℃	-	30	35	ms	Note1 Note3
View Angles		ΘT	CR≧10	80	85	-	Degree	Note 4
		ΘB		80	85	-		
		ΘL		80	85	-		
		ΘR		80	85	-		
Chromaticity	White	x	Brightness is on	Typ-0.05	0.290	Typ+0.05		Note5, Note1
		y			0.320			
	Red	x			TBD			
		y			TBD			
	Green	x			TBD			
		y			TBD			
	Blue	x			TBD			
		y			TBD			
Luminance		L		-	350	-	cd/m^2	Note1 Note6
Uniformity (White)		U		80	-	-	%	Note1 Note7

Environmental / Reliability Tests

No	Test Item	Condition	Judgment criteria
1	High Temp Operation	Ta= 70℃, 120hrs	Per table in below
2	Low Temp Operation	Ta= -20℃, 120hrs	Per table in below
3	High Temp Storage	Ts= 80℃, 120hrs	Per table in below
4	Low Temp Storage	Ts= -30℃, 120hrs	Per table in below
5	High Temp & High Humidity Storage	Ts= +60℃, 90% RH 120hours	Per table in below (polarizer discoloration is excluded)
6	Thermal Shock (Non-operation)	-30℃ 30 min~+80℃ 30 min, Change time:5min, 10 Cycles	Per table in below
7	ESD (Operation)	C=150pF, R=330Ω， 5points/panel Air:±8KV, 5times; Contact:±4KV, 5 times;	Per table in below
8	Vibration (Non-operation)	Frequency range:10~55Hz, Stroke:1.5mm Sweep:10Hz~55Hz~10Hz 2 hours for each direction of X.Y.Z.	Per table in below
9	Shock (Non-operation)	60G 6ms, ±X,±Y,±Z 3times, for each direction	Per table in below
10	Package Drop Test	Height:80 cm, 1 corner, 3 edges, 6 surfaces	Per table in below

INSPECTION	CRITERION(after test)
Appearance	No Crack on the FPC, on the LCD Panel
Alignment of LCD Panel	No Bubbles in the LCD Panel No other Defects of Alignment in Active area
Electrical current	Within device specifications
Function / Display	No Broken Circuit, No Short Circuit or No Black line No Other Defects of Display